20 mm 8 mm 2

Package Insert

AURO METFORMIN

Metformin Tablets BP 500 mg, 850 mg and 1000 mg

Rx Only

NAME OF DRUG PRODUCT:

Metformin Tablets BP 500 mg
Metformin Tablets BP 850 mg
Metformin Tablets BP 1000 mg

(TRADE) NAME OF PRODUCT:

AURO METFORMIN 500
AURO METFORMIN 850
AURO METFORMIN 1000

STRENGTH: 500 mg, 850 mg and 000 mg.

PHARMACEUTICAL DOSAGE FORM: Tablet

QUALITATIVE AND QUANTITATIVE COMPOSITIONS:

Metformin Tablets BP 500 mg
Each film-coated tablet contains:
Metformin Hydrochloride Ph. Eur..................500 mg

Metformin Tablets BP 850 mg
Each film-coated tablet contains:
Metformin Hydrochloride Ph. Eur.................. 850 mg

Metformin Tablets BP1000 mg
Each film-coated tablet contains:
Metformin Hydrochloride Ph. Eur…………..1000 mg

PHARMACEUTICAL FORM:

Metformin Tablets BP 500 mg: White, biconvex, circular shaped film coated tablets with 'A' debossed on one side and '60' debossed on the other side.

Metformin Tablets BP 850 mg: White, biconvex, circular shaped film coated tablets with 'A' debossed on one side and '61' debossed on the other side.

Metformin Tablets BP 1000 mg: White, biconvex, oval shaped film coated tablets with a score line in between '6' and '2' on one side and 'A' debossed on the other side.

CLINICAL PARTICULARS:

Pharmacodynamics/Pharmacokinetics

Pharmacodynamic properties

Metformin is a biguanide with anti hyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.

Metformin may act via 3 mechanisms:

(1) Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis.
(2) In muscle, by increasing insulin sensitivity, improving peripheral glucose uptake and utilization.
(3) Delay of intestinal glucose absorption.

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase.

Metformin increases the transport capacity of all types of membrane glucose transporters (GLUT).

In humans, independently of its action on glycaemia, Metformin has favourable effects on lipid metabolism. Metformin reduces total cholesterol, LDL cholesterol and triglyceride levels.

Pharmacokinetic properties

Absorption:

After an oral dose of Metformin, T_{max} is reached in 2.5 hours. Absolute bioavailability of a 500 mg or 850 mg Metformin tablet is approximately 50-60%. After an oral dose, the non-absorbed fraction recovered in faeces was 20-30%.

After oral administration, Metformin absorption is saturable and incomplete. It is assumed that the pharmacokinetics of Metformin absorption is non-linear. At the usual Metformin doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours and are generally less than 1 mcg/ml.

Food decreases the extent and slightly delays the absorption of Metformin. Following administration of a dose of 850 mg, a 40% lower plasma peak concentration, a 25% decrease in AUC (area under the curve) and a 35 minute prolongation of time to peak plasma concentration occurs.

Distribution:

Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean V_d ranged between 63-276 L.

Metabolism:

Metformin is excreted unchanged in the urine.

Elimination:

Renal clearance of Metformin is > 400 ml/min, indicating that Metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of Metformin in plasma.

Indication

Treatment of type 2 diabetes mellitus, particularly in overweight patients, when dietary management and exercise alone does not result in adequate glycaemic control.

In adults, Metformin film-coated tablets may be used as monotherapy or in combination with other oral anti-diabetic agents or with insulin.

In children from 10 years of age and adolescents, Metformin film-coated tablets may be used as monotherapy or in combination with insulin.

A reduction of diabetic complications has been shown in overweight type 2 diabetic adult patients treated with Metformin as first-line therapy after diet failure.

Recommended dose

Adults:

The usual starting dose of metformin hydrochloride tablets is 500 mg twice a day or 850 mg once a day, given with meals. Dosage increases should be made in increments of 500 mg weekly or 850 mg every 2 weeks, up to a total of 2000 mg per day, given in divided doses. Patients can also be titrated from 500 mg twice a day to 850 mg twice a day after 2 weeks. For those patients requiring additional glycemic control, Metformin may be given to a maximum daily dose of 3000 mg per day. Doses above 2000 mg may be better tolerated given three times a day with meals.

Monotherapy and combination with other oral anti diabetic agents:

The usual starting dose is one tablet 2 or 3 times daily given during or after meals.

After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose of Metformin is 3000 mg daily.

If transfer from another oral anti diabetic agent is intended: discontinue the other agent and initiate Metformin at the dose indicated above.

Combination with insulin:

Metformin and insulin may be used in combination therapy to achieve better blood glucose control. Metformin is given at the usual starting dose of one tablet 2-3 times daily, while insulin dosage is adjusted on the basis of blood glucose measurements.

Elderly: Due to the potential for decreased renal function in elderly, the Metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary.

Children and adolescents:

The usual starting dose of Metformin is 500 mg twice a day, given with meals. Dosage increases should be made in increments of 500 mg weekly up to a maximum of 2000 mg per day, given in divided doses.

Monotherapy and combination with insulin:

Metformin Tablets can be used in children from 10 years of age and adolescents. Metformin is not recommended in patients below the age of 10 years.

After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose of Metformin is 2 g daily, taken as 2 or 3 divided doses.

Pregnancy

Metformin is not recommended for use in pregnancy.

Renal impairment:

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3- 6 months.

GFR mL / min	Total maximum daily dose (to be divided into 2-3 daily doses)*	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated

** The text "to be divided into 2-3 daily doses" should be omitted for extended release products containing metformin as single agent.*

Contraindications

- Hypersensitivity to Metformin hydrochloride or to any of the excipients.
- Severely reduced kidney function (GFR <30 mL/min)

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A/s: 150 x 340 mm Black

AUROBINDO Packaging Development	Product Name: Auro-Metformin	Component: Leaflet	Item Code: P1514620	Date & Time: 05.03.2020 & 6:00 pm
	Customer / Country: Malaysia	Version No.: 10	Reason Of Issue: New	Reviewed / Approved by
Team Leader: Ganesh	Dimensions: 150 x 340 mm	No. of Colours : 01		
Initiator: C. Viswanath				
Artist: Vision Graphic Designers	Pharmacode: 14620	14620		Sign / Date
Additional Information :	Supersede Item code: P1506737			

- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis).
  Acute conditions with the potential to alter renal function such as dehydration, severe infection, shock, intravascular administration of iodinated contrast agents.
- Acute or chronic disease, which may cause tissue hypoxia such as cardiac or respiratory failure, recent myocardial infarction, shock.
- Hepatic insufficiency, acute alcohol intoxication, alcoholism.
- Lactation.

Warning and precautions

Lactic acidosis

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function

GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section Contraindications].

Children and adolescents:
The diagnosis of type 2 diabetes mellitus should be confirmed before treatment with Metformin is initiated. A careful follow-up of the effect of Metformin in metformin-treated children, especially pre-pubescent children, is recommended.

Children aged between 10 and 12 years:
Caution is recommended when prescribing to children aged between 10 and 12 years.

Other precautions:
- All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.

Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or sulphonylureas.

Interaction with other medicaments

Concomitant use not recommended:

Alcohol

Increased risk of lactic acidosis in acute alcohol intoxication, particularly in case of:
- fasting or malnutrition
- hepatic insufficiency

Avoid consumption of alcohol and alcohol-containing medications.

Iodinated contrast agents:

Intravascular administration of iodinated contrast agents may lead to renal failure, resulting in Metformin accumulation and a risk of lactic acidosis.

Metformin should be discontinued prior to or at the time of the test and not reinstituted until 48 hours afterwards and only after renal function has been re-evaluated and found to be normal.

Combinations requiring precautions for use:

Glucocorticoids (systemic and local routes), beta-2- agonists and diuretics have intrinsic hyper glycaemic activity. Inform the patient and perform more frequent blood glucose monitoring, especially at the beginning of treatment. If necessary, adjust the dosage of the anti diabetic drug during therapy with the other drug and upon its discontinuation.

ACE-inhibitors may decrease the blood glucose levels. If necessary, adjust the dosage of the anti diabetic drug during therapy with the other drug and upon its discontinuation.

Pregnancy and lactation

When the patient plans to become pregnant and during pregnancy, diabetes should not be treated with Metformin but insulin should be used to maintain blood glucose levels as close to normal as possible in order to lower the risk of foetal malformations associated with abnormal blood glucose levels.

Metformin is excreted into milk in lactating rats. Similar data are not available in humans and a decision should be made whether to discontinue nursing or to discontinue Metformin, taking into account the importance of the compound to the mother.

Effects on ability to drive and use machines

Metformin monotherapy does not cause hypoglycaemia and therefore has no effect on the ability to drive or to use machines. However, patients should be alerted to the risk of hypoglycaemia when Metformin is used in combination with other anti diabetic agents (sulphonylureas, insulin, repaglinide).

Side Effects

The following undesirable effects may occur under treatment with Metformin.

Metabolism and nutrition disorders:

Very rare: Decrease of vitamin B12 absorption with decrease of serum levels during long-term use of Metformin. Consideration of such aetiology is recommended if a patient presents with megaloplastic anaemia.

Very rare: Lactic acidosis

Nervous system disorders:

Common: Taste disturbance

Gastrointestinal disorders:

Very common: Gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. To prevent them, it is recommended that Metformin be taken in 2 or 3 daily doses during or after meals. A slow increase of the dose may also improveh gastrointestinal tolerability.

Skin and subcutaneous tissue disorders:

Very rare: Skin reactions such as erythema, pruritus, urticaria.

Symptoms and Treatment of Overdose

Hypoglycaemia may not be seen with Metformin doses of up to 85 g, although lactic acidosis may occur in such circumstances. High overdose or concomitant risks of Metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and Metformin is haemodialysis.

PHARMACEUTICAL PARTICULARS

List of excipients

Povidone, Magnesium Stearate, Hypromellose, Polyethylene glycol (400 & 6000) and Purified water.

Incompatibilities

None known.

Shelf life

60 months

Storage condition

Do not store above 30 °C.

Nature and contents of container

Blister pack:

AURO METFORMIN 500 & 850: Blister of 14 tablets.

AURO METFORMIN 1000: Blister of 10 tablets.

Manufactured by:

AUROBINDO

AUROBINDO PHARMA LTD.,
Unit –III, Sy. No. 313 & 314,
Bachupally, Bachupally Mandal,
Medchal-Malkajgiri District, Telangana State,
India

Regd. Office: Plot No.: 2, Maitrivihar
Ameerpet, Hyderabad-500 038
Telangana State, India.

Marketing Authorization holder in Malaysia:
Healol Pharmaceuticals Sdn Bhd,
74-3, Jalan Wangsa Delima 6, KLSC Wangsa Maju,
53300 Kuala Lumpur.

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